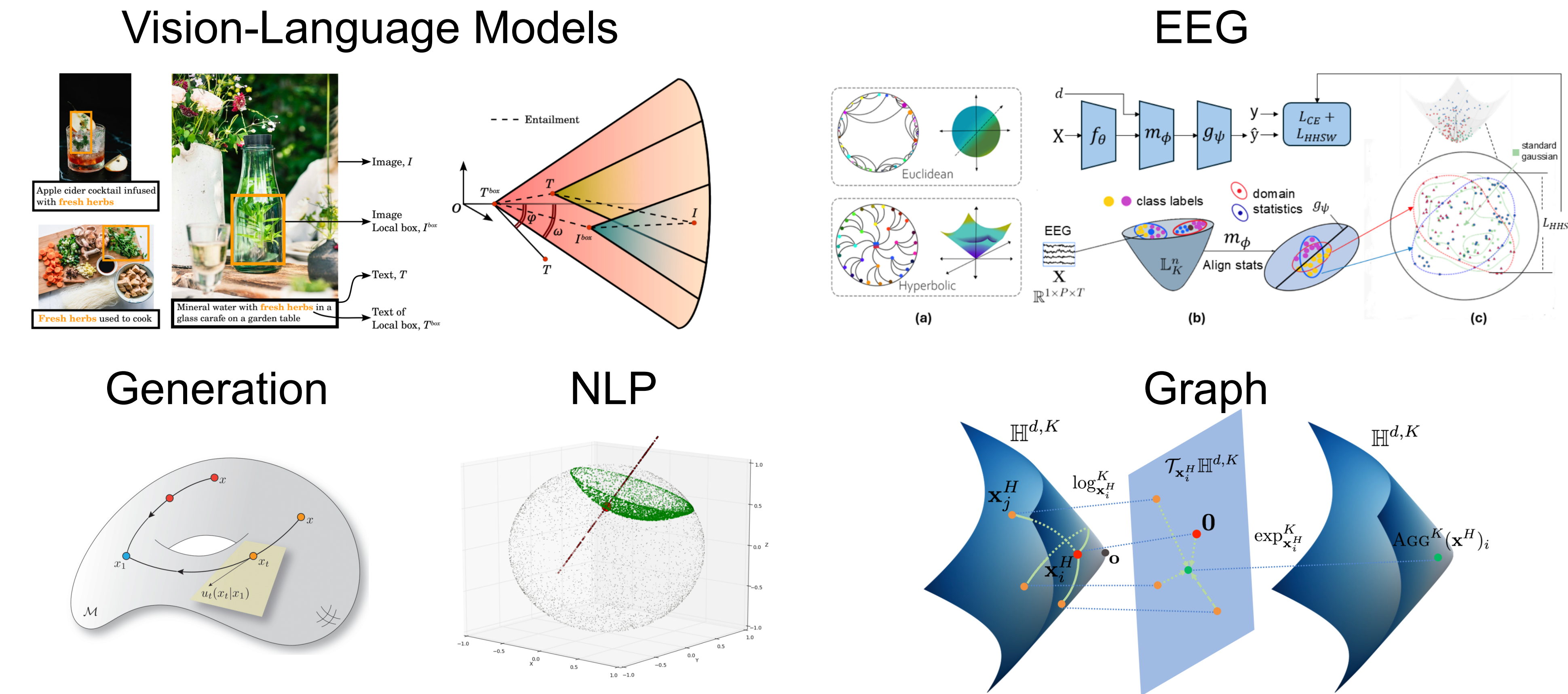




Motivation

Applications of Hyperbolic Spaces



Busemann and Horosphere

Busemann $\forall x \in \mathcal{H}_K^n, \quad B^\gamma(x) = \lim_{t \rightarrow \infty} (d(x, \gamma(t)) - t).$

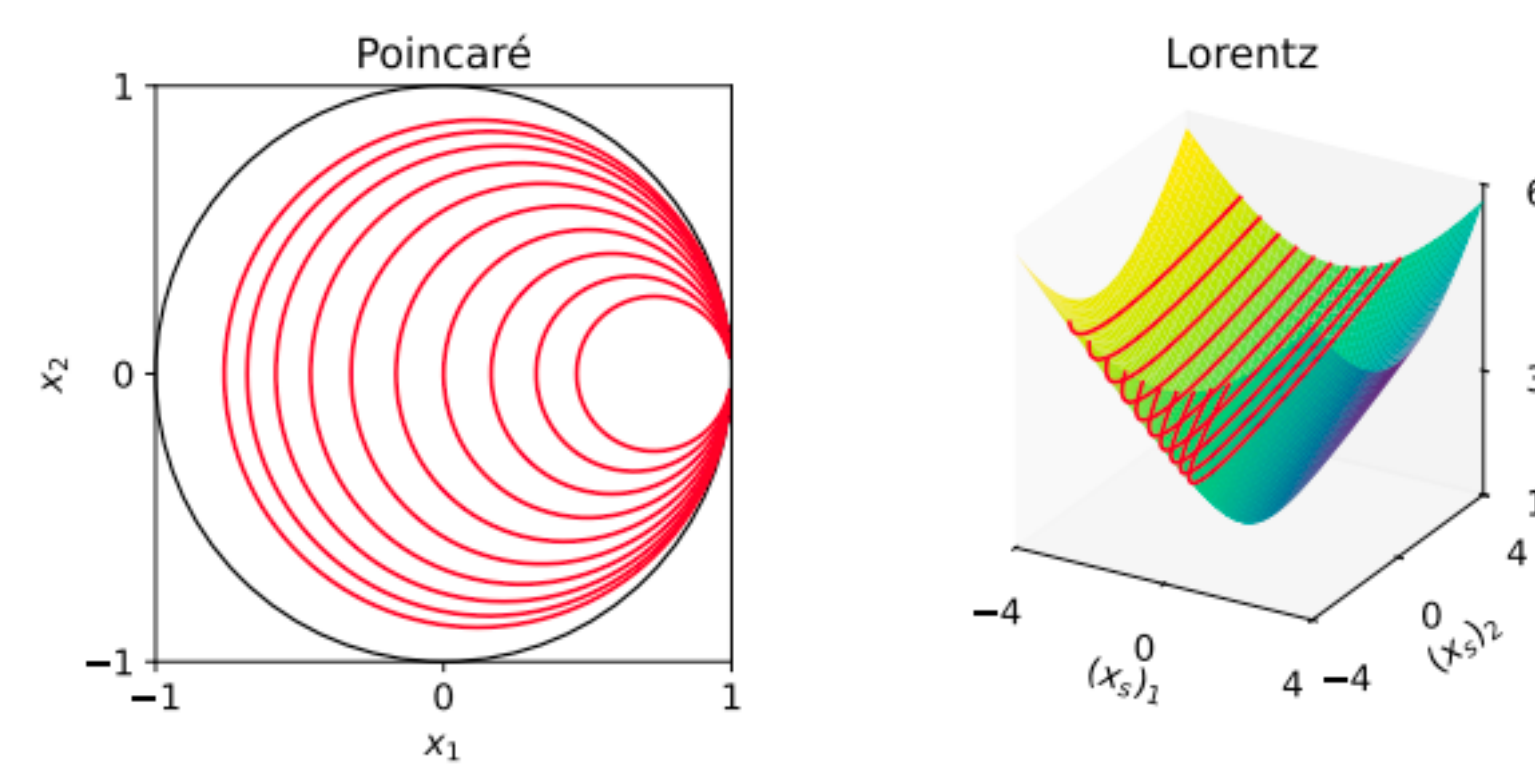
Horosphere $H_\tau^v = \{x \in \mathcal{H}_K^n \mid B^v(x) = \tau\}$

Example

$\forall x \in \mathbb{R}^n, \quad B^v(x) = -\langle x, v \rangle, \quad v \in \mathbb{S}^{n-1} = \{v \in \mathbb{R}^n \mid \|v\| = 1\}$

$$\mathbb{P}_K^n: B^v(x) = \frac{1}{\sqrt{-K}} \log \left(\frac{\|v - \sqrt{-K}x\|^2}{1 + K\|x\|^2} \right),$$

$$\mathbb{L}_K^n: B^v(x) = \frac{1}{\sqrt{-K}} \log \left(\sqrt{-K} (x_t - \langle x_s, v \rangle) \right).$$



Generalization

Table 1. Generalization: Euclidean vs. hyperbolic.

Euclidean	Hyperbolic
Straight line	Geodesic
Parallel lines	Asymptotic geodesic rays
Inner product $\langle v, x \rangle$	Busemann function $-B^v(x)$
Hyperplane	Horosphere
Parallel hyperplanes with $v \in \mathbb{S}^{n-1}$	Horospheres with $v \in \mathbb{S}^{n-1} \cap T_e \mathcal{H}_K^n$

Neural Network Layers

MLR

$$\forall k, p(y = k \mid x) = \frac{\exp(\langle a_k, x \rangle + b_k)}{\sum_{j=1}^C \exp(\langle a_j, x \rangle + b_j)},$$

$$u_k(x) = \alpha_k \langle v_k, x \rangle + b_k.$$

$$v_k = \frac{a_k}{\|a_k\|} \in \mathbb{S}^{n-1} \quad \alpha_k = \|a_k\| > 0$$

$$d(x, H_{a_k, b_k}) = \frac{|\langle a_k, x \rangle + b_k|}{\|a_k\|}$$

$$H_{a_k, b_k} = \{x \in \mathbb{R}^n : \langle a_k, x \rangle + b_k = 0\}$$

$$u_k(x) = \text{sign}(\langle a_k, x \rangle + b_k) \|a_k\| d(x, H_{a_k, b_k}),$$

FC

$$y = Ax + b \in \mathbb{R}^m$$

$$y_k = \langle a_k, x \rangle + b_k$$

Poincare

Lorentz

Generalization

$$\forall k, p(y = k \mid x) = \frac{\exp(u_k(x))}{\sum_{j=1}^C \exp(u_j(x))},$$

$$u_k(x) = -\alpha_k B^{v_k}(x) + b_k,$$

with $\alpha_k > 0, v_k \in \mathbb{S}^{n-1}$, and $b_k \in \mathbb{R}$ as parameters.

$$d(x, H_{v, \alpha, b}) = \frac{|-\alpha B^v(x) + b|}{\alpha}.$$

$$H_{v, \alpha, b} = \{x \in \mathcal{H}_K^n \mid -\alpha B^v(x) + b = 0\},$$

$$u_k(x) = \text{sign}_k \alpha_k d(x, H_{v_k, \alpha_k, b_k})$$

$$\text{sign}_k = \text{sign}(-\alpha_k B^{v_k}(x) + b_k)$$

$$y = \frac{\omega}{1 + \sqrt{1 - K\|\omega\|^2}}, \quad \omega = \left[\frac{\sinh(\sqrt{-K}u_k(x))}{\sqrt{-K}} \right]_{k=1}^m,$$

$$y = \begin{bmatrix} y_t \\ y_s \end{bmatrix} = \begin{bmatrix} \sqrt{\frac{1}{-K} + \|y_s\|^2} \\ \frac{1}{\sqrt{-K}} \sinh(\sqrt{-K}u(x)) \end{bmatrix}$$

$$\bar{d}(y, H_{e_k, e}) = \phi(u_k(x)), \quad \forall 1 \leq k \leq m.$$

$$\mathbb{L}_K^n \ni x \mapsto y = \begin{bmatrix} \sqrt{\frac{1}{-K} + \|y_s\|^2} \\ \frac{1}{\sqrt{-K}} \sinh(\sqrt{-K}u(x)) \end{bmatrix} \oplus_{\mathbb{L}} b \in \mathbb{L}_K^m,$$

Comparison

Method	Logit $u_k(x), \forall k \in \{1, \dots, C\}$	Space	Dist	#Params	Compact params	FLOPs	Batch efficiency
Euclidean MLR	$\langle a_k, x \rangle + b_k$, with $a_k \in \mathbb{R}^n, b_k \in \mathbb{R}$	$\mathbb{R}^n$	Real	$C(n+1)$	✓	$C(2n)$	✓
Poincaré MLR [23, Eq. (25)]	$\frac{\lambda_{p_k}^K \ a_k\ }{\sqrt{-K}} \sinh^{-1} \left(\frac{2\sqrt{-K} \langle -p_k \oplus_M x, a_k \rangle}{(1 + K \ -p_k \oplus_M x\ ^2) \ a_k\ } \right)$, with $p_k \in \mathbb{P}_K^n, a_k \in T_{p_k} \mathbb{P}_K^n$	$\mathbb{P}_K^n$	Real	$C(2n)$	✗	$C(19n + 29)$	✗
Poincaré MLR [51, Eq. (6)]	$\frac{2}{\sqrt{-K}} \alpha_k \sinh^{-1}(\alpha - \beta)$, $\alpha = \lambda_{p_k}^K \sqrt{-K} \langle x, v_k \rangle \cosh(2\sqrt{-K}b_k)$, $\beta = (\lambda_{p_k}^K - 1) \sinh(2\sqrt{-K}b_k)$, with $\alpha_k > 0, v_k \in \mathbb{S}^{n-1}, b_k \in \mathbb{R}$	$\mathbb{P}_K^n$	Real	$C(n+2)$	✓	$C(4n + 52)$	✓
Pseudo-Busemann MLR [45, Cor. 4.3]	$-d(x, p_k) \frac{B^{v_k}(-p_k \oplus_M x)}{\ -p_k \oplus_M x\ }$, with $p_k \in \mathbb{P}_K^n, v_k \in \mathbb{S}^{n-1}$	$\mathbb{P}_K^n$	Pseudo	$C(2n)$	✗	$C(19n + 34)$	✗
Lorentz MLR [3, Eq. (12)]	$\frac{1}{\sqrt{-K}} \text{sign}(\alpha) \beta \left \sinh^{-1} \left(\sqrt{-K} \frac{\alpha}{\beta} \right) \right $, $\alpha = \cosh(\sqrt{-K}b_k) \langle z_k, x_s \rangle - \sinh(\sqrt{-K}b_k)$, $\beta = \sqrt{\ \cosh(\sqrt{-K}b_k) z_k\ ^2 - (\sinh(\sqrt{-K}b_k) \ z_k\)^2}$, with $z_k \in \mathbb{R}^n, b_k \in \mathbb{R}$	$\mathbb{L}_K^n$	Real	$C(n+1)$	✓	$C(4n + 52)$	✓
BMLR	$-\alpha_k B^{v_k}(x) + b_k$, with $\alpha_k > 0, v_k \in \mathbb{S}^{n-1}, b_k \in \mathbb{R}$	$\mathbb{P}_K^n$ $\mathbb{L}_K^n$	Real	$C(n+2)$	✓	$\mathbb{P}_K^n: C(6n+12)$ $\mathbb{L}_K^n: C(2n+12)$	✓

Method	$\mathcal{F}: \mathcal{H}_K^n \ni x \mapsto y \in \mathcal{H}_K^m$	Space	Methodology	Parameters	#Params	FLOPs
Möbius [23, Eq. 27]	$\frac{1}{\sqrt{-K}} \tanh \left(\frac{\ Wx\ }{\ x\ } \tanh^{-1}(\sqrt{-K}\ x\) \right) \frac{Wx}{\ Wx\ }$	$\mathbb{P}_K^n$	Tangent	$W \in \mathbb{R}^{m \times n}$	mn	$2nm + 2n + 2m + 24$
Poincaré FC [51, Eq. (7)]	$y = \frac{\omega}{1 + \sqrt{1 - K\ \omega\ ^2}}, \omega_k = \frac{\sinh(\sqrt{-K}u_k(x))}{\sqrt{-K}}$, with $u_k(x)$ in Tab. 2	$\mathbb{P}_K^n$	Poincaré geometry	$\alpha_k > 0, v_k \in \mathbb{S}^{n-1}, b_k \in \mathbb{R}$, for $k = 1, \dots, m$	$m(n+2)$	$4nm + 71m + 4$
Lorentz FC [11, Eq. (3)]	$y = \begin{bmatrix} \sqrt{\ \psi(Wx, v)\ ^2 - 1/K} \\ \psi(Wx, v) \end{bmatrix}$, $\psi(Wx, v) = \lambda \sigma(v^\top x + b') \frac{W\phi(x) + b}{\ W\phi(x) + b\ }$ with ϕ and σ as the activation and sigmoid	$\mathbb{L}_K^n$	Ambient Minkowski	$W \in \mathbb{R}^{m \times (n+1)}, v \in \mathbb{R}^{n+1}, b \in \mathbb{R}^m, b' \in \mathbb{R}, \lambda > 0$	$m(n+1) + m + (n+1) + 2$	$2nm + 8m + 2n + 10$
BFC	$y = \frac{\omega}{1 + \sqrt{1 - K\ \omega\ ^2}}, \omega = \frac{\sinh(\sqrt{-K}u(x))}{\sqrt{-K}}, y_s = \frac{1}{\sqrt{-K}} \sinh(\sqrt{-K}u(x)), y_t = \sqrt{\frac{1}{-K} + \ y_s\ ^2}$, with $u_k(x) = \phi(-\alpha_k B^{v_k}(x) + b_k)$	$\mathbb{P}_K^n$ $\mathbb{L}_K^n$	Busemann	$\alpha_k > 0, v_k \in \mathbb{S}^{n-1}, b_k \in \mathbb{R}$, for $k = 1, \dots, m$	$m(n+2)$	$6nm + 29m + 4$ $2nm + 30m + 2$

Experiments

Image

Genome

Graph

FC in Graph

Method	CIFAR-10 (Num. classes: 10)			CIFAR-100 (Num. classes: 100)			Tiny-ImageNet (Num. classes: 200)			ImageNet-1k (Num. classes: 1000)		
	Acc	Fit Time	#Params	Acc	Fit Time	#Params	Acc	Fit Time	#Params	Acc	Fit Time	#Params
MLR	95.14 ± 0.12	10.66	5.13K	77.72 ± 0.15	10.60	51.30K	65.19 ± 0.12	69.17	102.60K	71.87	2263.12	513K
PMLR	95.04 ± 0.13	11.94	5.14K	77.19 ± 0.50	12.11	51.40K	64.93 ± 0.38	71.90	102.80K	71.77	2300.11	514K
PBMLR-P	95.23 ± 0.08	21.92	10.24K	77.78 ± 0.15	76.84	102.40K	65.43 ± 0.27	336.58	204.80K	71.46	3907.12	1024K
BMLR-P	95.32 ± 0.14	12.01	5.14K	78.10 ± 0.35	12.13	51.40K	66.16 ± 0.19	71.98	102.80K	73.36	2300.77	514K
LMLR	94.98 ± 0.12	11.55	5.13K	78.03 ± 0.21	11.72	51.30K	65.63 ± 0.10	69.27	102.60K	72.46	2277.17	513K
BMLR-L	95.25 ± 0.02	11.08	5.14K	78.07 ± 0.26	11.22	51.40K	65.99 ± 0.14	69.19	102.80K	73.24	2276.53	514K

Benchmark	Task	Dataset	Num. classes	$\mathbb{P}_K^n$			$\mathbb{L}_K^n$	
				PMLR	PBMLR-P	BMLR	LMLR	BMLR-L
TEB	Retrotransposons	LTR Copia	2	75.34 ± 1.02	74.37 ± 1.48	76.73 ± 1.08	73.01 ± 1.07	75.86 ± 1.52
		LINES	2	85.54 ± 0.61	85.92 ± 0.65	86.05 ± 1.08	83.14 ± 0.80	86.72 ± 0.58
		SINES	2	95.30 ± 0.85	95.34 ± 1.58	95.99 ± 0.74	96.70 ± 0.87	96.29 ± 0.59
	DNA transposons	CMC-EnSpn	2	83.39 ± 0.56	83.62 ± 1.00	84.03 ± 0.71	81.78 ± 1.05	84.15 ± 1.00
		hAT-Ac	2	89.38 ± 0.90	89.86 ± 0.54	89.62 ± 0.74	88.94 ± 0.69	90.70 ± 0.51
		processed unprocessed	2	72.45 ± 1.49	71.99 ± 2.04	73.09 ± 1.66	73.71 ± 1.76	73.32 ± 1.65
Pseudogenes	tata notata all	2	75.37 ± 2.27	71.99 ± 1.47	75.71 ± 1.89	74.54 ± 1.98	76.15 ± 1.61	
	Core Promoter Detection	tata notata all	2	80.95 ± 1.47	79.32 ± 2.44	80.29 ± 1.63	80.90 ± 1.15	81.76 ± 1.16
		tata notata all	2	70.02 ± 0.52	70.60 ± 0.75	70.48 ± 0.35	71.26 ± 0.56	70.43 ± 0.39
tata notata all		2	67.64 ± 0.77	68.02 ± 0.63	68.50 ± 0.61	67.63 ± 0.56	68.36 ± 1.07	
GUE	Promoter Detection	tata notata all	2	80.30 ± 1.59	80.27 ± 2.71	82.83 ± 1.69	83.27 ± 1.95	82.55 ± 1.54
		tata notata all	2	92.63 ± 0.36	93.05 ± 0.32	92.75 ± 0.51	91.74 ± 0.57	92.60 ± 0.49
		tata notata all	2	90.53 ± 0.50	90.79 ± 0.77	90.20 ± 0.65	89.34 ± 0.40	89.82 ± 0.45
Covid Variant Classification	Covid	9	74.09 ± 0.25	70.84 ± 0.80	73.40 ± 0.30	64.07 ± 0.51	72.45 ± 0.21	
Species Classification	Virus	20	67.24 ± 2.10	59.17 ± 3.32	77.12 ± 1.23	71.34 ± 2.05	77.21 ± 1.04	
	Fungi	25	15.06 ± 1.32	18.75 ± 1.77	30.01 ± 0.76	15.07 ± 1.88	30.14 ± 2.48	

Space	Method	Disease $\delta = 0$	Airport $\delta = 1$	PubMed $\delta = 3.5$	Cora $\delta = 11$
$\mathbb{P}_K^n$	HGCN	86.87 ± 2.58	85.34 ± 1.16	76.29 ± 0.98	76.56 ± 0.81
	HGCN-PMLR	88.98 ± 1.96	84.78 ± 1.48	76.02 ± 1.09	77.47 ± 1.15
	HGCN-PBMLR-P	89.05 ± 0.78	85.04 ± 0.97	75.89 ± 0.78	77.90 ± 1.00
	HGCN-BMLR-P	92.45 ± 0.96	86.02 ± 0.53	77.36 ± 0.73	78.48 ± 1.52
$\mathbb{L}_K^n$	HGCN	87.83 ± 0.77	84.94 ± 1.40	76.49 ± 0.88	77.37 ± 1.72
	HGCN-LMLR	89.72 ± 1.51	82.61 ± 1.01	75.44 ± 1.17	69.91 ± 3.61
	HGCN-BMLR-L	90.80 ± 1.15	85.27 ± 1.17	77.30 ± 0.41	77.65 ± 2.10

Table 7. Comparison of hyperbolic FC layers on link prediction. Best results within each hyperbolic model are in **bold**.

Space	Method	Methodology	Disease $\delta = 0$	Airport $\delta = 1$	PubMed $\delta = 3.5$	Cora $\delta = 11$
$\mathbb{P}_K^n$	Möbius	Tangent	76.35 ± 1.83	93.31 ± 0.41	94.93 ± 0.06	90.80 ± 0.56
	Poincaré FC	Poincaré geometry	79.45 ± 1.01	94.31 ± 0.16	94.24 ± 0.25	88.21 ± 0.72
	BFC-P	Busemann	80.45 ± 0.93	94.88 ± 0.39	94.85 ± 0.07	91.94 ± 0.32
$\mathbb{L}_K^n$	LTFC	Tangent	71.32 ± 5.36	92.68 ± 0.35	94.85 ± 0.17	89.37 ± 0.64
	Lorentz FC	Ambient Minkowski	72.78 ± 2.04	92.99 ± 0.33	94.20 ± 0.10	92.06 ± 0.62
	BFC-L	Busemann	78.36 ± 0.51	95.37 ± 0.17	94.90 ± 0.04	92.28 ± 0.12